Model answers of 2nd week Assessment

Group (A) – Model Answers

1- Arrange the following steps in the correct order:

1. The candle is lit.
2. The candle is placed inside a closed container.
3. The amount of oxygen decreases.
4. The candle flame goes out.

2- Give reason for each of the following:

a. The percentage of oxygen decreases during burning.

Because oxygen is used up during burning.

b. Burning candles are placed in open cups rather than closed containers.

Because fresh air, which contains oxygen, can reach the candle and support burning.

3- Complete the following:

a. The candle will keep burning in container number (1), because it contains oxygen.

b. The gas necessary for burning is oxygen.

4- Compare between:

a. The composition of air before and after burning

Before burning: air contains about 21% oxygen.

After burning: the percentage of oxygen decreases because oxygen is used in burning and the percentage of carbon dioxide increases.

b. The presence and absence of oxygen in the burning process

Oxygen present: burning continues.

Oxygen absent: burning stops or the candle flame goes out.

5- What would happen if ...?

a. Fresh air could not reach the fire

The flame would go out when the available oxygen is used up.

b. The amount of air around the candle increases

The candle may burn more strongly because more oxygen is available.

Group (B)- Model Answers

1- Choose the correct answer:

A. When burning occurs, the produced gas is:

(a) Oxygen (b) Carbon dioxide (c) Nitrogen (d) Hydrogen

Correct answer: (b) Carbon dioxide.

B. A burning candle inside a closed container, it is:

(a) burns more strongly. (b) keeps burning as it is.

(c) goes out after some time. (d) burns with a pop sound.

Correct answer: (c) goes out after some time.

2- Give a reason for each of the following:

A. Clear limewater is used to detect carbon dioxide gas.

Because carbon dioxide turns clear limewater milky.

B. The amount of oxygen decreases during burning.

Because oxygen is used up during the burning process.

3- Complete the following:

A. The candle flame goes out when all the oxygen gas is used up.

B. Carbon dioxide gas does not support burning.

4- Compare between:

A. The role of oxygen gas and carbon dioxide gas in the burning process.

Oxygen: supports burning. Carbon dioxide: does not support burning and can put out flames.

B. The composition of air before and after burning.

Before burning: air contains about 21% oxygen. After burning: the oxygen percentage decreases and carbon dioxide increases.

5- Answer the following questions:

A. Why does the candle not keep burning for a long time inside a sealed container?

Because the limited oxygen inside the sealed container is used up, so the flame goes out.

B. How can you know that air changes after burning?

The oxygen percentage decreases, and carbon dioxide is produced; carbon dioxide turns clear limewater milky.

C. What is the importance of oxygen in the burning process?

Oxygen is necessary for burning; without it, the flame goes out.

Group (C) – Model Answers

1- Choose the correct answer:

a. The candle flame becomes weaker and goes out because of:

Correct answer: (B) A decrease in oxygen.

b. The gas whose percentage is decreased after burning is:

Correct answer: (A) Oxygen.

2- Give reason:

a- The gas that increases after burning is carbon dioxide.

Because carbon dioxide is produced during burning.

b- Oxygen is important in the burning process.

Because oxygen supports burning; without it, the flame goes out.

3- Complete the following:

a. Oxygen gas helps in burning.

b. Air contains several gases, such as nitrogen, oxygen, and carbon dioxide.

4- Do all the gases that make up air help in burning? Explain your answer.

No. Oxygen supports burning, but gases such as nitrogen and carbon dioxide do not support burning.

5- Answer the following:

What would happen to the burning process if air did not flow?

The flame would become weaker and eventually go out because it would not receive a fresh supply of oxygen.